Reasoning with Uncertainty

Bayes' Theorem + Bayesian Networks

Past Paper Questions & Solutions (2020–2024)

CSE 713 Artificial Intelligence

Study Notes — June 2026

Contents

Exam Pattern Summary

Key Exam Facts

- **Yield:** 5/5 years, 4–9 marks per year (Group B, Q7 or Q8)
- **Pattern A — Bayes' Theorem:** compute $P(E)$, posteriors $P(H_i|E)$, most likely cause
- **Pattern B — BN joint distribution:** express $P(B, E, A, J, M)$, compute specific scenario
- **Recurring topology:** {Fire/Burglary, Earthquake} → Alarm → {Caller1, Caller2}
- **Key formula:** $P(H_i|E) = \dfrac{P(E|H_i) \cdot P(H_i)}{\sum_{k=1}^{K} P(E|H_k) \cdot P(H_k)}$
- **BN joint:** $P(X_1, \ldots, X_n) = \prod_i P(X_i|\text{Parents}(X_i))$

2022 WARNING — CPT values look wrong but are correct for that paper

2022 B-3d has P(A|F=T,E=T)=0.05 (alarm less likely when BOTH fire AND earthquake). This unusual CPT is verbatim from the exam paper. Use as given.
2023 Q7b uses P(F)=0.4, P(E)=0.6 (NOT 0.004/0.003 — those are 2022 values).

2024 — Q7 (9 marks total)

Q7a — Robot Vacuum Bayesian Network (4 marks)

2024 Q7a

A robot vacuum cleaner suddenly stops working (Evidence E). There are three possible hypotheses:

- H_1: Battery is low $P(H_1) = 0.5$, $P(E|H_1) = 0.7$
- H_2: Dust bin is full $P(H_2) = 0.3$, $P(E|H_2) = 0.8$
- H_3: Motor is overheated $P(H_3) = 0.2$, $P(E|H_3) = 0.9$

1. Compute $P(E)$, the total probability that the robot stops working.
2. Calculate the posterior probabilities $P(H_1|E)$, $P(H_2|E)$, $P(H_3|E)$.
3. Identify the most likely cause of stopping.
4. Justify your answer.

Solution

Step 1: Compute $P(E)$ using Total Probability:

$$P(E) = \sum_{i=1}^{3} P(E|H_i)\cdot P(H_i) = (0.7)(0.5)+(0.8)(0.3)+(0.9)(0.2) = 0.35+0.24+0.18 = \boxed{0.77}$$

Step 2: Compute posteriors using Extended Bayes' Theorem:

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{P(E)}$$

$$P(H_1|E) = \frac{0.7 \times 0.5}{0.77} = \frac{0.35}{0.77} \approx \boxed{0.4545}$$

$$P(H_2|E) = \frac{0.8 \times 0.3}{0.77} = \frac{0.24}{0.77} \approx \boxed{0.3117}$$

$$P(H_3|E) = \frac{0.9 \times 0.2}{0.77} = \frac{0.18}{0.77} \approx \boxed{0.2338}$$

Sanity check: $0.4545 + 0.3117 + 0.2338 = 1.000$ ✓

Step 3 & 4: Most likely cause and justification:

H_1 (Battery is low) is the most likely cause with posterior probability $\approx 45.45\%$. Although H_3 has the highest likelihood $P(E|H_3) = 0.9$, the strong prior $P(H_1) = 0.5$ makes H_1 dominate after combining both prior and likelihood via Bayes' theorem.

Q7b — Challenges in Bayesian Decision Theory (2 marks)

2024 Q7b

Identify and discuss the challenges in Bayesian decision theory.

Solution

Bayesian decision theory faces several key challenges:

1. **Prior elicitation:** Specifying accurate prior probability distributions is difficult. For n binary variables, a complete joint distribution requires 2^n entries — exponential growth. Expert elicitation is subjective and error-prone.
2. **Computational intractability:** Exact inference in general Bayesian Networks is NP-hard. Inference scales poorly with network size, necessitating approximate methods (MCMC, variational inference).
3. **Sensitivity to priors:** Results can vary significantly with different prior specifications, especially with limited training data. This subjectivity undermines objectivity of decisions.
4. **Conditional independence assumptions:** The BN structure encodes strong independence assumptions that may not hold in real-world domains, leading to incorrect probability estimates.
5. **Data requirements:** Learning accurate CPTs requires large amounts of training data. With sparse data, parameter estimates are unreliable.

Q7c — Evolutionary Method (3 marks)

2024 Q7c

Describe the evolutionary method with an appropriate example.

Solution

Evolutionary Algorithms (EAs) are population-based metaheuristic search methods inspired by biological evolution (natural selection, genetics).

Key Components:

- **Population:** A set of candidate solutions (individuals/chromosomes)
- **Fitness function:** Evaluates the quality of each individual (higher = better solution)
- **Selection:** Choose fitter individuals for reproduction (survival of the fittest)
- **Crossover:** Combine two parent chromosomes to produce offspring (exploration)
- **Mutation:** Random small changes to maintain diversity and avoid premature convergence

Example — Genetic Algorithm for Robot Path Planning:

- *Encoding:* Chromosome = sequence of waypoints, e.g. $[A, B, D, G]$

- *Fitness:* Inverse of total path length (shorter path $\rightarrow$ higher fitness)
- *Selection:* Tournament selection — compare 2 random individuals, keep the better
- *Crossover:* Single-point crossover of two path sequences to create new routes
- *Mutation:* Randomly swap two waypoints with small probability (e.g. 0.01)
- *Termination:* After N generations, output chromosome with highest fitness

After many generations, the algorithm converges to a near-optimal path through the environment.

2023 — Q7 (9 marks total)

Q7a — Three Bayes' Theorem Problems (3 marks)

2023 Q7a

Solve the following Bayes' Theorem problems:

(i) **Factory:** A factory has Machine A (produces 60% of items, 3% defect rate) and Machine B (produces 40%, 5% defect rate). A product is found defective. What is $P(\text{Machine A} \mid \text{defective})$?

(ii) **Medical Test:** A test is 98% accurate. Disease prevalence is 1%. A patient tests positive. What is $P(\text{disease} \mid \text{positive})$?

(iii) **Spam Filter:** Prior $P(\text{spam}) = 0.20$. $P(\text{“offer”} \mid \text{spam}) = 0.80$, $P(\text{“offer”} \mid \neg\text{spam}) = 0.10$. Find $P(\text{spam} \mid \text{contains “offer”})$.

Solution

(i) Factory Problem:
Let A = Machine A, B = Machine B, d = defective.

$P(d) = P(d|A){\cdot}P(A){+}P(d|B){\cdot}P(B) = (0.03)(0.6){+}(0.05)(0.4) = 0.018{+}0.020 = 0.038$

$$P(A \mid d) = \frac{P(d \mid A) \cdot P(A)}{P(d)} = \frac{0.018}{0.038} \approx \boxed{0.4737}$$

Probability the defective item came from Machine A $\approx 47.37\%$.

(ii) Medical Test:
$P(D) = 0.01$, $P(+|D) = 0.98$, $P(+|\neg D) = 1 - 0.98 = 0.02$ (false positive rate).

$$P(+) = (0.98)(0.01) + (0.02)(0.99) = 0.0098 + 0.0198 = 0.0296$$

$$P(D \mid +) = \frac{P(+|D) \cdot P(D)}{P(+)} = \frac{0.0098}{0.0296} \approx \boxed{0.3311}$$

Only $\approx 33.1\%$ chance of actually having disease despite positive test — base rate matters!

(iii) Spam Filter:

$$P(\text{“offer”}) = (0.80)(0.20) + (0.10)(0.80) = 0.16 + 0.08 = 0.24$$

$$P(\text{spam} \mid \text{“offer”}) = \frac{(0.80)(0.20)}{0.24} = \frac{0.16}{0.24} = \frac{2}{3} \approx \boxed{0.6667}$$

Q7b — Alarm Bayesian Network (6 marks)

CORRECTED Values for 2023 Q7b

The 2023 paper uses P(F)=**0.4**, P(E)=**0.6**. The values 0.004/0.003 belong to the 2022 paper (Yakin/Samin network). Verified from original exam paper.

2023 Q7b

A Bayesian Network models a home alarm system. Nodes: Fire (F), Earthquake (E), Alarm (A), Mr. X calls (X), Mr. Y calls (Y). The network topology is: $F \to A \leftarrow E$, $A \to X$, $A \to Y$.

Given probabilities:

- $P(F) = 0.4, \quad P(E) = 0.6$
- $P(A|F,E)$: $\{TT : 0.95,\ TF : 0.90,\ FT : 0.60,\ FF : 0.01\}$
- $P(X|A)$: $\{T : 0.80,\ F : 0.10\}$, $\quad P(Y|A)$: $\{T : 0.70,\ F : 0.20\}$

(i) Express the full joint distribution $P(F, E, A, X, Y)$.

(ii) Compute $P(A = T,\ F = F,\ E = F,\ X = T,\ Y = T)$.

(iii) Compute $P(A = T,\ F = T,\ E = F,\ X = T,\ Y = T)$.

Solution

(i) Joint Distribution (Chain Rule of BN):

$$P(F, E, A, X, Y) = P(F) \cdot P(E) \cdot P(A \mid F, E) \cdot P(X \mid A) \cdot P(Y \mid A)$$

(ii) Alarm sounds, no fire, no earthquake, both X and Y call:

$$\begin{aligned}
&P(A{=}T,\ F{=}F,\ E{=}F,\ X{=}T,\ Y{=}T)\\
&= P(F{=}F) \cdot P(E{=}F) \cdot P(A{=}T|F{=}F, E{=}F) \cdot P(X{=}T|A{=}T) \cdot P(Y{=}T|A{=}T)\\
&= (1 - 0.4) \cdot (1 - 0.6) \cdot 0.01 \cdot 0.80 \cdot 0.70\\
&= 0.6 \times 0.4 \times 0.01 \times 0.80 \times 0.70\\
&= 0.24 \times 0.01 \times 0.56 = \boxed{0.001344}
\end{aligned}$$

(iii) Alarm sounds, fire occurred, no earthquake, both X and Y call:

$$\begin{aligned}
&P(A{=}T,\ F{=}T,\ E{=}F,\ X{=}T,\ Y{=}T)\\
&= P(F{=}T) \cdot P(E{=}F) \cdot P(A{=}T|F{=}T, E{=}F) \cdot P(X{=}T|A{=}T) \cdot P(Y{=}T|A{=}T)\\
&= 0.4 \times 0.4 \times 0.90 \times 0.80 \times 0.70\\
&= 0.16 \times 0.90 \times 0.56 = 0.16 \times 0.504 = \boxed{0.08064}
\end{aligned}$$

2022 — B-3d (3 marks)

2022 B-3d — Alarm Network (Yakin & Samin)

A Bayesian Network: Fire(F), Earthquake(E), Alarm(A), Yakin calls(Y), Samin calls(S).
Topology: $F \to A \leftarrow E$, $A \to Y$, $A \to S$.

Given:

- $P(F) = 0.004, \quad P(E) = 0.003$
- $P(A|F,E)$: $\{TT: 0.05,\ TF: 0.94,\ FT: 0.29,\ FF: 0.001\}$
- $P(Y|A)$: $\{T: 0.90,\ F: 0.05\}, \quad P(S|A)$: $\{T: 0.80,\ F: 0.01\}$

(i) Express the joint probability distribution $P(F, E, A, Y, S)$.

(ii) Compute $P(A = T,\ F = F,\ E = F,\ Y = T,\ S = T)$.

(iii) Compute $P(A = T,\ F = T,\ E = F,\ Y = T,\ S = T)$.

Solution

(i) Joint Distribution:

$$P(F, E, A, Y, S) = P(F) \cdot P(E) \cdot P(A \mid F, E) \cdot P(Y \mid A) \cdot P(S \mid A)$$

(ii) Alarm sounds, no fire, no earthquake, both Y and S call:

$$\begin{aligned}
&P(A{=}T,\ F{=}F,\ E{=}F,\ Y{=}T,\ S{=}T)\\
&= (1 - 0.004) \cdot (1 - 0.003) \cdot P(A{=}T|FF) \cdot P(Y{=}T|A{=}T) \cdot P(S{=}T|A{=}T)\\
&= 0.996 \times 0.997 \times 0.001 \times 0.90 \times 0.80\\
&= 0.993012 \times 0.001 \times 0.72 = 0.993012 \times 0.00072 \approx \boxed{7.15 \times 10^{-4}}
\end{aligned}$$

(iii) Alarm sounds, fire occurred, no earthquake, both Y and S call:

$$\begin{aligned}
&P(A{=}T,\ F{=}T,\ E{=}F,\ Y{=}T,\ S{=}T)\\
&= 0.004 \times (1 - 0.003) \times P(A{=}T|TF) \times P(Y{=}T|A{=}T) \times P(S{=}T|A{=}T)\\
&= 0.004 \times 0.997 \times 0.94 \times 0.90 \times 0.80\\
&= 0.003988 \times 0.94 \times 0.72 = 0.003749 \times 0.72 \approx \boxed{2.70 \times 10^{-3}}
\end{aligned}$$

2021 — Q7 (8.75 marks total)

Q7a — Concept of Uncertainty (2.75 marks)

2021 Q7a

Explain the concept of uncertainty using the doorbell-at-midnight example. The doorbell rang at 12 o'clock midnight.

- Was someone at the door?
- Did Karim wake up?

Propositions: Prop1: AtDoor(x) $\rightarrow$ Doorbell; Prop2: Doorbell $\rightarrow$ Wake(Karim)

Solution

Question 1 — Was someone at the door?
This requires abductive reasoning: since Doorbell occurred, conclude AtDoor(x) via Prop1.
Problem: Prop1 is **incomplete**. The doorbell might ring due to:

- Electrical malfunction, wind pressure, an animal, a faulty switch
- The list of possible causes is effectively **infinite**

Conclusion: We **cannot** conclude with certainty that someone was at the door.

Question 2 — Did Karim wake up?
This requires deductive reasoning: Doorbell $\rightarrow$ Wake(Karim) via Prop2.
Problem: Prop2 is **not a tautology**. Karim may not wake up if:

- He is in deep sleep, wearing earplugs, or is away from home

Conclusion: We **cannot** conclude with certainty that Karim woke up.

Sources of uncertainty demonstrated:

- *Incomplete propositions* (Prop1 has infinite possible antecedents)
- *Non-tautological implications* (Prop2 holds "often" but not always)
- *Uncertain conclusions* from both abductive and deductive reasoning

The solution: model uncertainty explicitly using probabilities (Bayesian approach).

Q7b — Why Deductive & Abductive Reasoning Are Not Sound (2.5 marks)

2021 Q7b

Explain why deductive and abductive (adductive) reasoning are not sound. What is the source of uncertainty in such reasoning?

Solution

Deductive reasoning is not always sound because:
The KB (knowledge base) is a *restricted, simplified model* of the real world. Even if

the deduction is formally valid within the model, it may be wrong in reality because:

- The model may be incomplete or incorrect
- Propositions assumed universally true may have exceptions
- Example: "Doorbell $\rightarrow$ Wake(Karim)" leads to a deductively valid conclusion, but the real-world proposition has exceptions (deep sleep, absent, etc.)

Abductive reasoning is not sound because:
It commits the logical fallacy of **affirming the consequent**:

$$P \rightarrow Q, \quad \text{observed } Q \quad \therefore P$$

This is logically invalid because many different causes can produce the same observation.

- Example: Many things can cause the doorbell to ring, not only someone at the door

Source of uncertainty:

1. Implications may be weak (not universally true)
2. Propositions use imprecise language ("often", "usually", "rarely")
3. Knowledge representation is a restricted model of reality
4. Inference process itself introduces uncertainty (deductive results may be formally correct but real-world-wrong; abductive results are not logically guaranteed)

Q7c — Meningitis: Bayes' Theorem (3.5 marks)

2021 Q7c

A doctor knows that meningitis causes a patient to have a stiff neck 40% of the time. The doctor also knows that:

$$P(\text{meningitis}) = \frac{1}{50000}, \qquad P(\text{stiff neck}) = \frac{1}{25}$$

What is the probability that a patient with a stiff neck has meningitis?

Solution

Let M = meningitis, S = stiff neck.
Given: $P(S|M) = 0.40$, $P(M) = \frac{1}{50000}$, $P(S) = \frac{1}{25}$
Applying Bayes' Theorem:

$$P(M \mid S) = \frac{P(S \mid M) \cdot P(M)}{P(S)} = \frac{0.40 \times \dfrac{1}{50000}}{\dfrac{1}{25}} = 0.40 \times \frac{25}{50000} = \frac{0.40}{2000} = \boxed{\frac{1}{5000} = 0.0002}$$

Interpretation: Even though stiff neck is a notable symptom of meningitis (40% chance if you have meningitis), the disease is so rare (1 in 50000) that even given the symptom, the probability of actually having meningitis is only **0.02%**. This illustrates the critical importance of **base rate (prior) in Bayesian reasoning**.

2021 — Q8a — Burglary Alarm Network (3.5 marks)

2021 Q8a

Consider a Bayesian Network with nodes: Burglary (B), Earthquake (E), Alarm (A), JohnCalls (J), MaryCalls (M). Topology: $B \to A \leftarrow E$, $A \to J$, $A \to M$.

Given probabilities:

- $P(B) = 0.001, \quad P(E) = 0.002$
- $P(A|B,E)$: $\{TT : 0.95,\ TF : 0.94,\ FT : 0.29,\ FF : 0.001\}$
- $P(J|A)$: $\{T : 0.90,\ F : 0.05\}$, $\quad P(M|A)$: $\{T : 0.70,\ F : 0.01\}$

(i) Express the joint probability distribution $P(B, E, A, J, M)$.

(ii) Compute $P(A = T,\ B = F,\ E = F,\ J = T,\ M = T)$.

(iii) Compute $P(A = T,\ B = T,\ E = F,\ J = T,\ M = T)$.

Solution

(i) Joint Distribution:

$$P(B, E, A, J, M) = P(B) \cdot P(E) \cdot P(A \mid B, E) \cdot P(J \mid A) \cdot P(M \mid A)$$

(ii) Alarm sounds, no burglary, no earthquake, both John and Mary call:

$$\begin{aligned}
&P(A{=}T,\ B{=}F,\ E{=}F,\ J{=}T,\ M{=}T)\\
&= (1 - 0.001) \cdot (1 - 0.002) \cdot P(A{=}T|FF) \cdot P(J{=}T|A{=}T) \cdot P(M{=}T|A{=}T)\\
&= 0.999 \times 0.998 \times 0.001 \times 0.90 \times 0.70\\
&= 0.997002 \times 0.001 \times 0.63\\
&= 0.000997002 \times 0.63 \approx \boxed{6.28 \times 10^{-4}}
\end{aligned}$$

(iii) Alarm sounds, burglary occurred, no earthquake, both John and Mary call:

$$\begin{aligned}
&P(A{=}T,\ B{=}T,\ E{=}F,\ J{=}T,\ M{=}T)\\
&= 0.001 \times (1 - 0.002) \times P(A{=}T|TF) \times P(J{=}T|A{=}T) \times P(M{=}T|A{=}T)\\
&= 0.001 \times 0.998 \times 0.94 \times 0.90 \times 0.70\\
&= 0.000998 \times 0.94 \times 0.63\\
&= 0.00093812 \times 0.63 \approx \boxed{5.91 \times 10^{-4}}
\end{aligned}$$

2020 — Q7 (Group B)

Q7a — Uncertainty Concept (Doorbell) (≈1.5 marks)

2020 Q7a

Explain the concept of uncertainty using an appropriate example. [Same doorbell scenario as 2021 Q7a]

Solution

See 2021 Q7a solution above (identical concept). The doorbell-at-midnight example illustrates:

- Both abductive reasoning (someone at door?) and deductive reasoning (Karim woke?) fail under uncertainty
- Propositions that are "often true but not tautologies" create uncertainty
- Solution: model uncertainty using probability theory (Bayesian approach)

Q7d — Extension of Bayes' Theorem (3 marks)

2020 Q7d

Describe the extension of Bayes' theorem. Give the equations and explain the semantics of two extended theorems. Name the application areas.

Solution

Extended Bayes' Theorem — Form 1 (Multiple Hypotheses):

$$\boxed{P(H_i \mid E) = \frac{P(E \mid H_i) \cdot P(H_i)}{\sum_{k=1}^{K} P(E \mid H_k) \cdot P(H_k)}}$$

Semantics of Form 1:

- $P(H_i|E)$: *posterior* — probability hypothesis H_i is true given evidence E
- $P(E|H_i)$: *likelihood* — probability of observing E if H_i is true
- $P(H_i)$: *prior* — a priori probability of H_i without specific evidence
- K: total number of mutually exclusive, exhaustive hypotheses
- Denominator $= P(E)$ — total probability of observing evidence E across all hypotheses

Extended Bayes' Theorem — Form 2 (Updating with New Evidence):

$$\boxed{P(H \mid E, e) = \frac{P(H \mid E) \cdot P(e \mid E, H)}{P(e \mid E)}}$$

Semantics of Form 2:

- Updates belief in H given existing evidence E when *new* evidence e arrives
- $P(H|E)$ plays the role of the new prior (posterior from previous update)
- Allows **sequential/incremental Bayesian updating** — each new piece of evidence refines the estimate
- The size of joint probabilities needed grows as 2^n for n propositions

Application Areas:

1. **Medical diagnosis:** $P(\text{disease} \mid \text{symptoms})$ — differential diagnosis
2. **Spam filtering:** $P(\text{spam} \mid \text{email features})$ — Naive Bayes classifier
3. **Robot localization:** $P(\text{position} \mid \text{sensor readings})$ — particle filters
4. **Weather forecasting:** Update rain probability as new atmospheric data arrives
5. **Fault diagnosis:** Industrial systems — identify machine failure from sensor data

2020 — Q8 (Group B)

Q8a — Bayesian Network Syntax and Semantics (1.5 marks)

2020 Q8a

Explain the syntax and semantics of a Bayesian Network.

Solution

Syntax — Structure of a Bayesian Network:
A Bayesian Network $N = (X, G, P)$ consists of:

- A **DAG** $G = (V, E)$: directed acyclic graph with nodes $V = \{v_1, \ldots, v_n\}$ (random variables) and directed edges E (causal influences)
- A set of **random variables** X represented by the nodes of G
- A set of **Conditional Probability Tables (CPT)**: one distribution $P(X_v \mid X_{pa(v)})$ for each variable X_v, conditioned on its parents $pa(v)$
- *Root nodes* (no parents) carry **prior probability** tables; all others carry CPTs

Semantics — What a BN Represents:
A BN encodes a compact **factorization of the full joint probability distribution**:

$$\boxed{P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))}$$

Markov Condition: Each node is *conditionally independent* of all its non-descendants given its parents. This is the independence assumption that makes BNs tractable.
Example (5-node alarm network):

$$P(B, E, A, J, M) = P(B) \cdot P(E) \cdot P(A|B, E) \cdot P(J|A) \cdot P(M|A)$$

Q8b — Meningitis (same as 2021 Q7c) (1.5 marks)

2020 Q8b

Using Bayes' theorem: $P(M) = \frac{1}{50000}$, $P(S|M) = 0.40$, $P(S) = \frac{1}{25}$. Find $P(M|S)$.

Solution

$$P(M \mid S) = \frac{P(S \mid M) \cdot P(M)}{P(S)} = \frac{0.40 \times \frac{1}{50000}}{\frac{1}{25}} = 0.40 \times \frac{25}{50000} = \frac{10}{50000} = \boxed{0.0002}$$

Only 0.02% — very low due to the extremely small prior $P(M) = 1/50000$.

Q8c — Complex BN: Burglary, Earthquake, Reliable & Unreliable Friend (≈6.75 marks)

2020 Q8c

A Bayesian Network models a burglar alarm system. Nodes: Burglary (B), Earthquake (E), Alarm (A), ReliableFriend (R), UnreliableFriend (C).

Topology: $B, E \to A$; $A \to R, C$.

Given:

- $P(B) = 0.02$, $P(E) = 0.001$
- $P(A|\neg B, \neg E) = 0.01$; $P(A|B, \neg E) = 0.9$; $P(A|\neg B, E) = 0.9$; $P(A|B, E) = 0.99$
- $P(R|A) = 0.002$; $P(R|\neg A) = 0.999$; $P(C|A) = 0.8$; $P(C|\neg A) = 0.1$

(i) Draw the Bayesian Network.

(ii) Compute prior $P(\text{alarm is sounding}) = P(A = T)$.

(iii) Compute the normalized likelihood $P(A \mid R = T, C = T)$.

(iv) Compute the posterior probability.

(v) How would you modify the BN if an earthquake radio report is also available?

Solution

(i) Network Structure:

$$B \longrightarrow A \longleftarrow E \qquad A \longrightarrow R \qquad A \longrightarrow C$$

(5-node DAG: B and E are roots; A has parents B and E; R and C are leaves)

(ii) Prior $P(A = T)$ using Total Probability:

$$\begin{aligned}
P(A) &= P(A|B, E)P(B)P(E) + P(A|B, \neg E)P(B)P(\neg E) \\
&\quad + P(A|\neg B, E)P(\neg B)P(E) + P(A|\neg B, \neg E)P(\neg B)P(\neg E) \\
&= 0.99(0.02)(0.001) + 0.9(0.02)(0.999) + 0.9(0.98)(0.001) + 0.01(0.98)(0.999) \\
&= 0.0000198 + 0.017982 + 0.000882 + 0.0097902 \\
&\approx \boxed{0.02867}
\end{aligned}$$

(iii) Normalized likelihood $P(A|R = T, C = T)$:
Note: $P(R|A) = 0.002$ is unusual (reliable friend rarely signals when alarm sounds

in this model). Use as given.

$$\begin{aligned} P(A{=}T, R{=}T, C{=}T) &= P(A{=}T) \cdot P(R{=}T|A{=}T) \cdot P(C{=}T|A{=}T) \\ &= 0.02867 \times 0.002 \times 0.8 = 4.587 \times 10^{-5} \end{aligned}$$

$$\begin{aligned} P(A{=}F, R{=}T, C{=}T) &= P(A{=}F) \cdot P(R{=}T|A{=}F) \cdot P(C{=}T|A{=}F) \\ &= 0.97133 \times 0.999 \times 0.1 \approx 0.09704 \end{aligned}$$

$$P(A{=}T|R{=}T, C{=}T) = \frac{4.587 \times 10^{-5}}{4.587 \times 10^{-5} + 0.09704} \approx \boxed{0.000473}$$

(iv) Posterior $P(B = T \mid A = T)$:

$P(A|B) = P(A|B, E)P(E)+P(A|B, \neg E)P(\neg E) = 0.99(0.001)+0.9(0.999) = 0.00099+0.8991 = 0.90$

$$P(B|A{=}T) = \frac{P(A{=}T|B) \cdot P(B)}{P(A{=}T)} = \frac{0.90009 \times 0.02}{0.02867} \approx \frac{0.018002}{0.02867} \approx \boxed{0.628}$$

(v) Modified BN for Earthquake Radio Report:
Add a new node **RadioReport** as an additional child of Earthquake:

- $E \to$ RadioReport: $P(\text{Report}|E = T) \approx 0.9$; $P(\text{Report}|E = F) \approx 0.05$

The modified topology becomes: $B, E \to A \to R, C$; and also $E \to$ RadioReport. When RadioReport is observed, it provides additional evidence for/against Earthquake, which then propagates to update $P(A)$.

Quick Reference Summary

Core Formulas to Memorise

Formula	Description
$P(H_i\|E) = \dfrac{P(E\|H_i)P(H_i)}{\sum_k P(E\|H_k)P(H_k)}$	Extended Bayes (multiple hypotheses)
$P(H\|E,e) = \dfrac{P(H\|E)\cdot P(e\|E,H)}{P(e\|E)}$	Sequential update form
$P(x_1,\ldots,x_n) = \prod_i P(x_i\|\mathrm{Pa}(x_i))$	BN joint factorization (chain rule)
$P(E) = \sum_i P(E\|H_i)\cdot P(H_i)$	Total probability (marginalization)

Numeric Results to Memorise

Problem	Key Result
2024 Q7a: Robot vacuum P(E)	0.77; Most likely H_1 (battery) $\approx 45.45\%$
2023 Q7a(i): Factory defect	$P(\text{MachineA}\|\text{defect}) \approx 0.4737$
2023 Q7a(ii): Medical test	$P(\text{disease}\|+) \approx 0.3311$ (only 33%!)
2023 Q7a(iii): Spam filter	$P(\text{spam}\|\text{“offer”}) = 2/3 \approx 0.667$
2023 Q7b(i): BN scenario	$P \approx 0.001344$
2023 Q7b(ii): BN scenario	$P \approx 0.08064$
2021 Q7c / 2020 Q8b: Meningitis	$P(M\|S) = 0.0002 = 0.02\%$
2021 Q8a(ii): BN no burglary	$\approx 6.28 \times 10^{-4}$
2021 Q8a(iii): BN burglary	$\approx 5.91 \times 10^{-4}$
2022 B-3d(ii): No fire/eq	$\approx 7.15 \times 10^{-4}$
2022 B-3d(iii): Fire only	$\approx 2.70 \times 10^{-3}$

BN Inference Types (4 Patterns)

1. **Diagnostic** (effects $\rightarrow$ causes): Given John calls, find $P(\text{Burglary})$
2. **Causal** (causes $\rightarrow$ effects): Given Burglary, find $P(\text{John calls})$
3. **Inter-causal** (between causes of common effect): Given Alarm, find $P(\text{Burglary}|\text{also Earthquake})$
4. **Mixed**: Given John calls and no Earthquake, find $P(\text{Alarm})$

Common Mistakes in Exams

- **Wrong 2023 values:** 2023 Q7b uses P(F)=0.4, P(E)=0.6 (NOT 0.004/0.003). The 0.004/0.003 values are from 2022 B-3d.
- **Forgetting to check:** $\sum_i P(H_i|E) = 1$ after computing all posteriors
- **Joint distribution factorization:** Must multiply ALL factors — prior nodes $\times$ CPT nodes $\times$ leaf nodes
- **Meningitis base rate:** $P(M|S) = 0.0002$, NOT 0.40 (the 40% is $P(S|M)$, not the answer)
- **BN chain rule:** The joint = product of $P(\text{node}|\text{parents})$ for EVERY node